

BACSWN SWOT and Risk Analysis

BACSWN — SWOT & Risk Analysis

Document: BACSWN-RISK-2026-008 **Classification:** CONFIDENTIAL — For Authorized Recipients Only **Prepared for:** Owner & CEO, BACSWN (private delivery partner) **Prepared by:** Sky Miles Limited — AI Elevate Division **Version:** 1.0 — June 2026 **Companion to:** Master Framework Schedule (BACSWN-SCHED-2026-003); Mesh Architecture Master Plan (BACSWN-MARCH-2026-002)

1. Purpose

This document gives the owner/CEO a single strategic view of BACSWN's **strengths, weaknesses, opportunities, and threats (SWOT)** and the **consolidated, living risk register** for the implementation program. It is the authoritative risk artifact (maintained under WS-A — Program & Governance) and supersedes the risk fragments embedded in the schedule, which it consolidates and expands.

2. SWOT Analysis

Strengths

- **Platform already built and live** — in production, security-reviewed (15/15), 137/137 tests; the hardest technical step is behind us.
- **Secured PPP (Build-Own-Operate)** — sole-source national infrastructure; no competitive bake-off to lose, and the partner owns and operates the asset.
- **Autonomous, sovereign mesh** — stations keep sensing, predicting, and warning even when the hub or an island is cut off; resilient exactly where centralized systems fail.
- **Engineered survivability** — Cat 4/5 hardening with a satellite lifeline; link budgets confirm the design (Architecture Annex A).
- **Defense-grade security** — full end-to-end cryptography (hardware root-of-trust, PKI, FIPS-grade, post-quantum path).
- **One feed, three missions** — a single surveillance layer drives safety, overflight-fee accounting, and CORSIA carbon — efficient and hard to replicate.
- **National-response backbone** — deep emergency-services integration (NEMA, Police, Fire/Rescue, RBDF, shelters) via a shared Common Operating Picture.
- **User-funded model** — recurring revenue tied to airline overflight/landing fees over busy oceanic airspace (a diversified, hard-currency demand base).

Weaknesses

- **Remote-island O&M** — the most critical sites (Inagua, Mayaguana, Ragged Island) are the hardest and costliest to reach; sustainment is the program's hardest ongoing task.
- **Long-lead, ITAR-controlled radar** — the Raytheon primary-radar program is a dependency with export licensing and heavy civil works.
- **External-feed reliance** — Iridium satellite and Tomorrow.io are third-party (mitigated as supplementary, but real).
- **Edge-hardware obsolescence** — Jetson-class compute evolves fast over a long BOO life.
- **VHF horizon limits** — some hops exceed practical line-of-sight and depend on relay/satellite.
- **Initial local skills gap** — Bahamian operating capacity must be built (addressed by BACSWN-TRN-2026-007).
- **Program complexity** — 18 parallel workstreams across 24 months is significant execution surface.

Opportunities

- **Regional replication** — the platform/IP could template to other hurricane-exposed Caribbean and small-island states (optional future upside).
- **Research & academic partnerships** — a distributed edge-AI weather network attracts collaboration and talent.
- **Technology transfer** — mesh, consensus, and edge-AI methods have wider application.
- **Climate-resilience funding** — the safety/impact case attracts grants and development partners.
- **National capability & leadership** — positions the Bahamas as a regional aviation-weather leader.
- **Adjacent services** — value-added products on the airspace, weather, and emissions data.

Threats

- **Major hurricane during deployment** — a strike on a partially-built network.
- **Air-traffic volume shock** — pandemic-type downturn cutting the fee-based revenue.
- **Regulatory / ITAR / spectrum delay** — URCA spectrum and US radar export licensing on the critical path.

- **Political / mandate continuity** — administration change or re-scoping.
- **Cyber or physical attack** — critical national infrastructure is a target.
- **Supply-chain / vendor disruption** — edge hardware, radar, satellite.
- **Environmental constraints** — protected-area issues forcing site changes.
- **Untrusted AI supply chain** — Chinese-built/untrusted model weights (excluded by policy; see R20).

3. Risk Assessment Method

Each risk is scored **Likelihood (L) × Impact (I)** on a 1–5 scale; severity bands: **1–6 Low, 8–12 Medium, 15–25 High/Critical**. Residual = rating after the stated mitigation. Owners map to schedule workstreams (WS-A...R).

4. Consolidated Risk Register

ID	Category	Risk	L×I	Severity	Mitigation	Residual	Owner
R01	Force majeure	Hurricane strikes partially-deployed network	4×5	Critical	South-to-north sequencing; satellite fallback live before June 1; pilot/remote sites first	Medium	WS-A/C
R02	Force majeure	In-service major hurricane damages live sites	3×4	High	Cat 4/5 hardening; satellite survival core; pre-positioned spares; post-storm drone-first assessment	Medium	WS-G/I
R03	Regulatory	VHF spectrum (URCA) licensing delay	3×4	High	File M1; satellite/cellular interim path	Low	WS-B
R04	Regulatory	Raytheon radar — ITAR/export delay or install complexity	4×4	Critical	File ITAR + order at M1; parallel civil works; phased install; ADS-B + Tomorrow.io interim coverage	Medium	WS-P/B
R05	Environmental	Land rights / permitting / protected-area delay (gates civil works)	3×4	High	Start M0; engage authorities early; prioritize pilot+radar sites; hold re-siting options	Low	WS-Q
R06	Commercial / demand	Air-traffic volume downturn cuts overflight/landing-fee revenue	3×4	High	Diversified airline base; minimum-traffic floor in concession; opex flexibility	Medium	WS-A
R07	Commercial	Fee collection / remittance via civil-aviation authority	2×3	Medium	Contractual fee-share & remittance terms; reconciliation in metering (REV-1)	Low	WS-A/L
R08	Security	Cyber or physical attack on critical infrastructure	3×5	Critical	End-to-end crypto (\$4.4); SecOps/IR; pen-testing; tamper-zeroization; physical site security	Medium	WS-J
R09	Technical	Edge-AI model underperformance	3×3	Medium	Hub-based fallback models; phased validation per gate	Low	WS-F
R10	Supply chain	Edge-hardware supply disruption	3×3	Medium	Pre-order; qualified second sources (Pi 5 + Coral, AMD/Xilinx)	Low	WS-D
R11	Technical / lifecycle	Edge-hardware obsolescence over BOO life	3×3	Medium	Refresh roadmap; modular FRU design; second sources	Medium	WS-D/I
R12	Technical	VHF horizon limits leave coverage gaps	2×3	Medium	Multi-hop routing; satellite overlay; link-budget verification at SAT (Annex A)	Low	WS-C
R13	Vendor	Dependency / lock-in (Raytheon, Iridium, Tomorrow.io)	3×3	Medium	Treat external feeds as supplementary; support contracts; avoid sole-source critical paths	Medium	WS-A/L
R14	Regulatory	U.S./military data-sharing agreement delayed (military track fusion)	3×3	Medium	Commercial ADS-B first; pursue MOU early; primary-radar fallback	Low	WS-L/O
R15	Regulatory	Drone/UAS approval (BCAA)	2×3	Medium	Early BCAA engagement; use own air picture for deconfliction	Low	WS-N
R16	Operational	Remote-island O&M access (cost, weather, logistics)	4×3	High	Regional service hubs; drone-assisted triage; predictive maintenance; satellite keeps sites reporting	Medium	WS-I
R17	Operational / people	Local skills gap / staffing for sovereign operation	3×3	Medium	Training & Capacity-Building Plan; train-the-trainer; academic partnerships; phased localization	Low	WS-R
R18	Political	Administration change / mandate continuity	2×4	Medium	BOO asset ownership; essential-service status; contractual term protections	Medium	WS-A
R19	Program	Execution complexity across 18 workstreams	3×3	Medium	Phase gates; PMO; critical-path focus; change control; consolidated risk reviews	Low	WS-A
R20	Security / supply chain	AI model provenance — untrusted or Chinese-built model	2×4	Medium	Policy: trusted, non-Chinese models only; provenance + integrity verification (Data Governance	Low	WS-J

ID	Category	Risk	LxI	Severity	Mitigation	Residual	Owner
		weights			§6); applies to edge ensemble and hub		

5. Top Risks (focus list)

The **critical-band** risks demanding continuous owner/CEO attention:

1. **R01 — Hurricane during deployment** (sequencing + satellite-first remote sites).
2. **R04 — Raytheon radar / ITAR** (longest defense lead time; file and order at M1).
3. **R08 — Cyber/physical attack** (security is a design baseline, not an add-on).

High-band watch list: **R03** (URCA), **R05** (permitting), **R06** (air-traffic demand), **R16** (remote O&M).

6. Risk Governance

- **Living register:** this document is maintained under WS-A; risks re-scored and residuals reviewed at the schedule's governance cadence (weekly RAG to owner/CEO; monthly review).
- **Escalation:** any risk moving into the critical band, or a new critical risk, is escalated to the owner/CEO immediately.
- **Gate linkage:** open critical/major risks block the relevant phase gate (per the V&V plan).
- **Contingency:** schedule float and re-siting/second-source options are held for the highest-impact dependencies (radar, permitting, supply chain).

See also: Master Framework Schedule §7 (schedule-impacting risks, now consolidated here); V&V Plan (BACSWN-VV-2026-006); O&M Plan (BACSWN-OM-2026-004); Training & Capacity-Building Plan (BACSWN-TRN-2026-007).